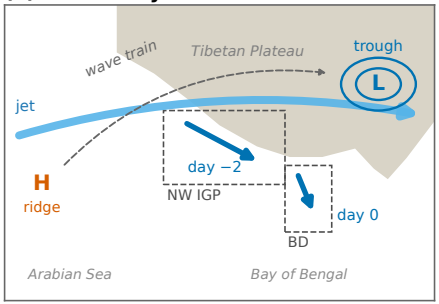
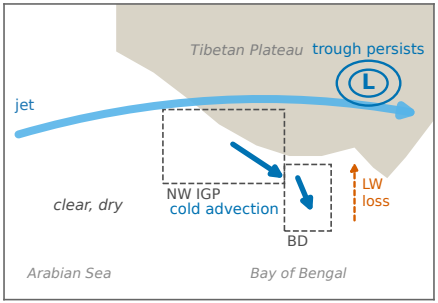
## (a) Onset: days −15 to 0



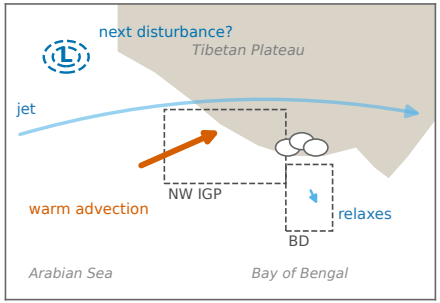

- Rossby wave train along the subtropical jet; trough over South Asia builds from day −13; jet +3.4 m $s^{-1}$ at onset
- cold advection: NW IGP on day −2 (−0.8 K $d^{-1}$), then BD on day 0 (−1.2 K $d^{-1}$)
- Tmin −2.4 °C at onset
- Siberian High, Ural blocking not anomalous on average

## (b) Maintenance




- sustained cold advection over BD (−1.2 to −1.4 K $d^{-1}$)
- dry, clear air: dewpoint −2.8 °C, cloud −0.11, longwave −21 W $m^{-2}$ → strong night-time cooling
- solar +12 W $m^{-2}$ → days recover first; DTR peaks on the last day
- duration largely undetected at onset

## (c) Termination




- falling heights, weaker jet: higher termination odds
- warm advection returns over NW IGP 2 and 1 days before the last day (+0.3, +0.6 K $d^{-1}$; lag-1 OR 1.83)
- next day over BD: cold advection relaxes to climatology; moisture and cloud return; Tmin rises by about 1.2 °C

